

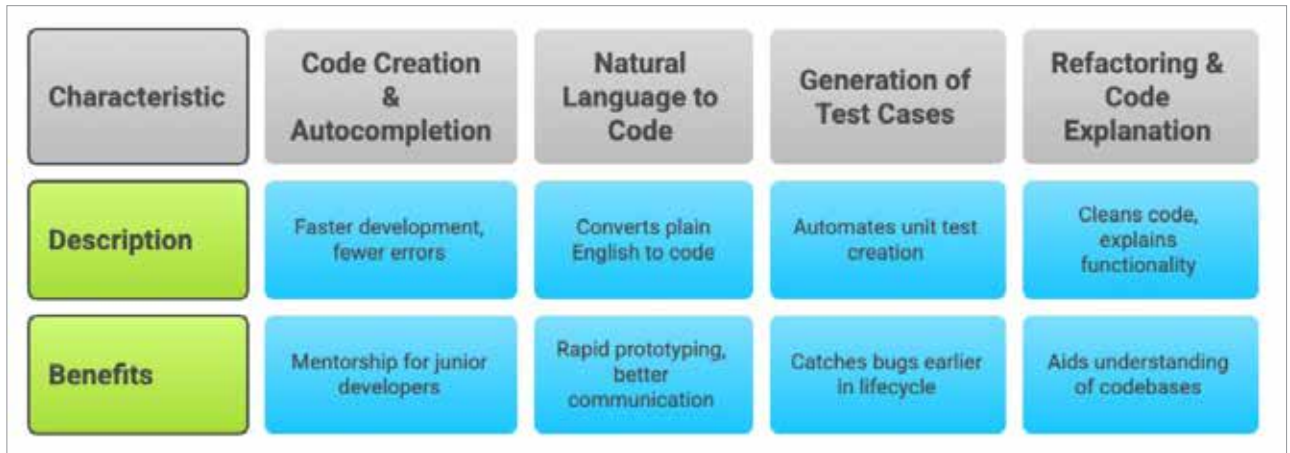


Figure 2: GenAI impact on programming

huge datasets of code and text, which make up their large language models (LLM), and prompt them to understand how to respond in advanced ways even better than experienced developers.

Generative AI represents a paradigm shift as it brings natural human language and computer code closer than ever before, leading to a future where coding becomes as much communication as logic.

How GenAI is transforming programming and the SDLC

Generative AI tools do not simply replace mundane tasks but redefine the process of building an application. Here is how these tools are helping.

Code creation and autocompletion:

Code generation has traditionally been a painstaking task that required deep knowledge of syntax, libraries, and logic structures. Generative AI tools such as GitHub Copilot and Tabnine change that by providing suggestions for complete lines or blocks of code as programmers type.

These tools are clearly more than just autocomplete; they know something about the context of your project and the structure of a given function, and they actually get the meaning of your comments. This means they allow for faster development in fewer lines of code and with less syntax errors. To junior developers, they offer

mentorship; for senior developers, they act as a speed multiplier.

Natural language to code: One of the most revolutionary aspects of GenAI is converting plain English into working code. Developers can now state the desired functionality, such as “write a function to sort a list of objects by a date field,” and the AI can generate the implementation in Python, JavaScript, or whichever language one chooses.

This encourages non-developers to participate, enables rapid prototyping, and enhances communication between the technical and non-technical. This also paves the way for a more creative and exploratory approach to software design.

Table 1: Popular Gen AI tools for developers

Tool	Developer	Key features	Best for
GitHub Copilot	GitHub + OpenAI	Autocompletes code, suggests functions, understands comments	General-purpose development, fast prototyping
Amazon CodeWhisperer	AWS	Code suggestions, security scans, AWS best practices	Cloud-native apps, AWS infrastructure automation
ChatGPT (w/ Code Interpreter)	OpenAI	Conversational code generation, explanations, debugging	Learning, multi-language support, code exploration
Tabnine	Tabnine Inc.	Local/private models, multi-language support, fast autocompletion	Teams prioritising privacy and on-prem solutions
Replit Ghostwriter	Replit	In-browser code suggestions, error fixes, and explanations	Students, beginners, real-time collaboration
Codeium	Ex-FAANG engineers	Free autocompletion for 70+ languages, IDE integration	Cost-effective alternative to Copilot
Cody (Sourcegraph)	Sourcegraph	Understands large codebases, answers natural language questions	Navigating and documenting enterprise-level projects